Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Charles Ahlem Ranch

Physical address of dairy:

Number and Street	City	County	Zip Code
23546 W American AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address): Washington

Date facility was originally placed in operation: 10/01/1982

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X130-X057-XXXX X045-X180-X037-XXXX

B. OPERATORS

Ahlem, Charles

Operator name: Ahlem, Charles Telephone no.: (209) 668-0867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

Ahlem, Mark

Operator name: Ahlem, Mark Telephone no.: (209) 668-0807 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95374

This operator is responsible for paying permit fees.

C. OWNERS

Ahlem, Charles

Legal owner name: Ahlem, Charles Telephone no.: (209) 668-0867 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95324

Ahlem, Mark

Legal owner name: Ahlem, Mark

Telephone no.: (209) 668-0807

Landline	Cellular

Mailing Address Number and Street	City	State	Zip Code
23546 American AVE	Hilmar	CA	95374

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	866	404	0
Number under roof	2,740	0	0	0	0	0
Maximum number	2,740	0	0	866	404	0
Average number	2,740	0	0	866	404	0
Avg live weight (lbs)	1,000	0	0	500		

Predominant milk cow breed: Jersey

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 76,081.03 *tons per reporting period*

Total nitrogen from manure: 979,725.07 *lbs per reporting period*

After ammonia losses (30% loss applied): 685,807.55 *lbs per reporting period*

Total phosphorus from manure: 163,208.31 *lbs per reporting period*

Total potassium from manure: 506,733.48 *lbs per reporting period*

Total salt from manure: 1,290,129.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 57,818,000 *gallons*

Total nitrogen generated: 142,681.35 *lbs*

Total phosphorus generated: 51,759.39 *lbs*

Total potassium generated: 1,017,872.15 *lbs*

Total salt generated: 3,603,430.75 *lbs*

	41,205,000	*gallons applied*
+	16,613,000	*gallons exported*
-	0	*gallons imported*
=	57,818,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/27/2023	Separator solids *Solid Manure-DC #6*	2,000.00 *ton*	Dry-weight	55.0	5,400.00	2,200.00	4,500.00		75.10
05/31/2023	Separator solids *Solid Manure-CMC Jerseys*	2,300.00 *ton*	Dry-weight	85.3	22,100.00	10,200.00	6,100.00		18.70

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/01/2023	Solid commercial fertilizer *CAN 17*	1.50 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000
06/01/2023	Solid commercial fertilizer *UN32*	1.75 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,628.37	0.00	0.00	0.00
Dry manure	24,664.02	10,857.24	12,224.82	1,478,249.40
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	26,292.39	10,857.24	12,224.82	1,478,249.40

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
01/06/2023	Process wastewater	390,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
01/30/2023	Process wastewater	690,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
02/02/2023	Process wastewater	180,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
02/15/2023	Process wastewater	225,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
03/03/2023	Process wastewater	540,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
03/10/2023	Process wastewater	735,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
04/10/2023	Process wastewater	1,290,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
04/16/2023	Process wastewater	360,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
04/26/2023	Process wastewater	380,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
06/04/2023	Process wastewater	600,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
06/10/2023	Process wastewater	780,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
06/18/2023	Process wastewater	240,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
06/28/2023	Process wastewater	720,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
07/08/2023	Process wastewater	420,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
07/27/2023	Process wastewater	690,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
08/05/2023	Process wastewater	1,710,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
08/07/2023	Process wastewater	405,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
08/12/2023	Process wastewater	300,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
08/13/2023	Process wastewater	680,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
08/15/2023	Process wastewater	1,800,000.00 *gal*	286.00	84.00	0.00	0.00	114.00	2,350.00		7,800
08/23/2023	Process wastewater	855,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
09/07/2023	Process wastewater	615,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
09/22/2023	Process wastewater	360,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
09/22/2023	Process wastewater	864,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
09/27/2023	Process wastewater	384,000.00 *gal*	143.00	98.00	0.00	0.00	61.90	750.00		3,780
11/02/2023	Process wastewater	400,000.00 *gal*	549.00	370.00	0.00	0.00	31.70	505.00		6,720

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	34,700.59	13,406.72	254,436.96	913,942.23
Total exports for all materials	34,700.59	13,406.72	254,436.96	913,942.23

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
50	50	50	2	both	X045-X180-X046-XXXX
70	65	65	2	both	X045-X020-X032-XXXX
Corn Shed	35	35	2	process wastewater	X045-X180-X037-XXXX X045-X180-X038-XXXX
Hennes	34	34	2	process wastewater	X045-X180-X001-XXXX
Horse Pasture	11	11	2	both	X045-X020-X032-XXXX
Manuel N/S	15	15	2	process wastewater	X045-X180-X008-XXXX
Manuel S/S	18	18	2	process wastewater	X045-X180-X008-XXXX
Marks Trees	11	11	1	none	X045-X320-X009-XXXX X045-X320-X039-XXXX
Morris	18	18	2	both	X045-X020-X020-XXXX
Pasture	34	34	2	both	X045-X040-X011-XXXX
Silveria 1	5	5	2	both	X045-X020-X032-XXXX
Silveria 2	23	23	2	both	X045-X020-X032-XXXX
The 60	65	65	2	both	X045-X180-X049-XXXX
Vitorino Field	20	20	0	none	X045-X180-X020-XXXX
Totals for areas that were used for application	373	373	24		
Totals for areas that were not used for application	31	31	1		
Land application area totals	404	404	25		

B. CROPS AND HARVESTS

50

Field name: 50

50

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	890.29 *ton*	As-is		71.1	5,200.00	1,100.00	7,700.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.81	185.18	39.17	274.21	1,317.34

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,477.96 *ton*	As-is		59.7	5,000.00	1,000.00	7,500.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.56	295.59	59.12	443.39	1,620.08

70

Field name: 70

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	912.96 *ton*	As-is		64.5	5,600.00	1,100.00	6,900.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.05	157.31	30.90	193.83	1,126.87

70

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	1,724.00 *ton*	As-is		51.0	4,900.00	500.00	5,900.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.52	259.93	26.52	312.97	1,299.63

Corn Shed

Field name: Corn Shed

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	643.67 *ton*	As-is		70.7	5,400.00	1,200.00	8,100.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.39	198.62	44.14	297.93	1,293.22

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	910.01 *ton*	As-is		62.0	5,200.00	1,400.00	10,700.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	270.40	72.80	556.41	1,679.62

Hennes

Field name: Hennes

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	495.88 *ton*	As-is		67.7	5,200.00	1,100.00	7,600.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.58	151.68	32.09	221.69	961.02

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	782.39 *ton*	As-is		62.0	5,200.00	1,400.00	10,700.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.01	239.32	64.43	492.45	1,486.54

Horse Pasture

Field name: Horse Pasture

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	190.02 *ton*	As-is		68.0	3,900.00	800.00	6,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.27	134.74	27.64	210.75	1,359.85

Horse Pasture

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	297.07 *ton*	As-is		59.9	4,300.00	800.00	5,700.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.01	232.25	43.21	307.87	1,277.89

Manuel N/S

Field name: Manuel N/S

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	258.20 *ton*	As-is		66.6	4,300.00	1,000.00	7,000.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.21	148.03	34.43	240.99	1,264.84

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	417.93 *ton*	As-is		63.7	3,900.00	600.00	5,200.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.86	217.32	33.43	289.76	1,517.09

Manuel S/S

Field name: Manuel S/S

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/15/2023	309.84 *ton*	As-is		64.7	5,300.00	1,200.00	8,200.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.21	182.46	41.31	282.30	1,336.79

06/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	506.69 *ton*	As-is		59.9	4,500.00	900.00	6,200.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.15	253.35	50.67	349.05	1,422.28

Marks Trees

Field name: Marks Trees

01/01/2023: Almonds

Crop: Almonds Acres planted: 11 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	15.20 *ton*	As-is		10.0	65,000.00	11,000.00	70,500.00		0.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	152.80	16.80	146.40	0.00
Total actual harvest content	1.38	179.64	30.40	194.84	0.00

Morris

Field name: Morris

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	268.69 *ton*	As-is		64.7	4,500.00	1,000.00	6,000.00		8.29

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.93	134.35	29.85	179.13	873.65

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	381.68 *ton*	As-is		58.7	5,700.00	800.00	6,200.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	21.20	241.73	33.93	262.94	1,173.50

Pasture

Field name: Pasture

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	566.94 *ton*	As-is		68.0	3,900.00	800.00	6,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.67	130.06	26.68	203.43	1,312.63

Pasture

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	833.00 *ton*	As-is		59.9	4,300.00	800.00	5,700.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.50	210.70	39.20	279.30	1,159.29

Silveria 1

Field name: Silveria 1

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 5 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	84.26 *ton*	As-is		67.3	5,100.00	1,200.00	7,900.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.85	171.89	40.44	266.26	1,245.40

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 5 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	137.00 *ton*	As-is		60.7	4,400.00	500.00	6,600.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.40	241.12	27.40	361.68	1,184.50

Silveria 2

Field name: Silveria 2

11/09/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/09/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	392.73 *ton*	As-is		67.3	5,100.00	1,200.00	7,900.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.08	174.17	40.98	269.79	1,261.89

06/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	619.00 *ton*	As-is		60.7	4,400.00	500.00	6,600.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.91	236.83	26.91	355.25	1,163.45

The 60

Field name: The 60

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/16/2023	1,249.75 *ton*	As-is		66.8	5,200.00	1,200.00	8,500.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.23	199.96	46.14	326.86	1,404.33

The 60

06/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 06/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,653.92 *ton*	As-is		62.2	4,900.00	800.00	6,100.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.44	249.36	40.71	310.43	1,385.02

NUTRIENT BUDGET

A. LAND APPLICATIONS

50 - 11/10/2022: Oats, silage-soft dough

Field name: 50

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.67	7,793,750.00 *gal*
Application event totals		0.00	0.00	0.00	279.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.10	13.93	129.71	1,331.76	990,000.00 *gal*
Application event totals		103.10	13.93	129.71	1,331.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.91	9.13	188.26	624.87	480,000.00 *gal*
Application event totals		22.91	9.13	188.26	624.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.64	11.42	235.33	781.09	600,000.00 *gal*
Application event totals		28.64	11.42	235.33	781.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.96	17.12	352.99	1,171.64	900,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	237.83	6,627,950.00 *gal*
Application event totals		42.96	17.12	352.99	1,409.47	

50 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	216.87	6,043,750.00 *gal*
Application event totals		0.00	0.00	0.00	216.87	

50 - 06/19/2023: Corn, silage

Field name: 50

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported SM	Separator solids	129.95	59.98	35.87	1,099.56	1,000.00 *ton*
Application event totals		129.95	59.98	35.87	1,099.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.50	12.56	258.86	859.20	660,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	299.71	8,352,200.00 *gal*
Application event totals		31.50	12.56	258.86	1,158.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	271.06	7,553,850.00 *gal*
Application event totals		0.00	0.00	0.00	271.06	

50 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.12	19.98	411.83	1,366.91	1,050,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.07	6,132,950.00 *gal*
Application event totals		50.12	19.98	411.83	1,586.98	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.42	21.69	447.13	1,484.07	1,140,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	196.48	5,475,550.00 *gal*
Application event totals		54.42	21.69	447.13	1,680.56	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	180.22	5,022,500.00 *gal*
Application event totals		0.00	0.00	0.00	180.22	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.22	13.24	272.98	906.07	696,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	135.44	3,774,350.00 *gal*
Application event totals		33.22	13.24	272.98	1,041.50	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	143.83	4,008,118.00 *gal*
Application event totals		0.00	0.00	0.00	143.83	

50 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	165.54	4,613,201.00 *gal*
Application event totals		0.00	0.00	0.00	165.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	175.55	4,892,113.00 *gal*
Application event totals		0.00	0.00	0.00	175.55	

70 - 11/07/2022: Oats, silage-soft dough

Field name: 70

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	1,430.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	266.55	9,656,790.00 *gal*
Application event totals		0.00	0.00	0.00	266.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	270.86	9,812,920.00 *gal*
Application event totals		0.00	0.00	0.00	270.86	

70 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	219.94	7,968,220.00 *gal*
Application event totals		0.00	0.00	0.00	219.94	

70 - 06/20/2023: Corn, silage

Field name: 70

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported SM	Separator solids	149.54	60.92	124.62	20,796.92	2,000.00 *ton*
Application event totals		149.54	60.92	124.62	20,796.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.61	13.00	267.91	889.24	888,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	289.27	10,479,625.00 *gal*
Application event totals		32.61	13.00	267.91	1,178.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	266.12	9,641,125.00 *gal*
Application event totals		0.00	0.00	0.00	266.12	

70 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.47	8.96	184.64	612.86	612,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	246.78	8,940,425.00 *gal*
Application event totals		22.47	8.96	184.64	859.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.47	19.32	398.25	1,321.85	1,320,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	223.92	8,112,325.00 *gal*
Application event totals		48.47	19.32	398.25	1,545.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	208.73	7,562,110.00 *gal*
Application event totals		0.00	0.00	0.00	208.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	217.76	7,889,260.00 *gal*
Application event totals		0.00	0.00	0.00	217.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.72	12.65	260.67	865.21	864,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	193.76	7,019,760.00 *gal*
Application event totals		31.72	12.65	260.67	1,058.97	

Corn Shed - 11/08/2022: Oats, silage-soft dough

Field name: Corn Shed

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Corn Shed - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.13	7.72	71.87	737.95	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	285.27	5,564,860.00 *gal*
Application event totals		57.13	7.72	71.87	1,023.21	

01/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.10	19.57	403.42	1,339.01	720,000.00 *gal*
Application event totals		49.10	19.57	403.42	1,339.01	

02/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	30.69	12.23	252.14	836.88	450,000.00 *gal*
Application event totals		30.69	12.23	252.14	836.88	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.28	15.66	322.74	1,071.21	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	243.66	4,753,280.00 *gal*
Application event totals		39.28	15.66	322.74	1,314.88	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.28	15.66	322.74	1,071.21	576,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	216.37	4,220,895.00 *gal*
Application event totals		39.28	15.66	322.74	1,287.58	

Corn Shed - 06/17/2023: Corn, silage

Field name: Corn Shed

Crop: Corn, silage

Plant date: 06/17/2023

Corn Shed - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.64	11.42	235.33	781.09	420,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	300.55	5,862,920.00 *gal*
Application event totals		28.64	11.42	235.33	1,081.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.50	8.97	184.90	613.72	330,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	268.50	5,237,750.00 *gal*
Application event totals		22.50	8.97	184.90	882.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.23	22.02	453.85	1,506.39	810,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	234.81	4,580,625.00 *gal*
Application event totals		55.23	22.02	453.85	1,741.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.65	29.36	605.13	2,008.52	1,080,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	222.69	4,344,200.00 *gal*
Application event totals		73.65	29.36	605.13	2,231.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.73	13.05	268.95	892.68	480,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	212.62	4,147,675.00 *gal*
Application event totals		32.73	13.05	268.95	1,105.30	

Corn Shed - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.91	9.13	188.26	624.87	336,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	193.03	3,765,615.00 *gal*
Application event totals		22.91	9.13	188.26	817.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.73	13.05	268.95	892.68	480,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	143.97	2,808,430.00 *gal*
Application event totals		32.73	13.05	268.95	1,036.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	184.86	3,606,130.00 *gal*
Application event totals		0.00	0.00	0.00	184.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.19	10.44	215.16	714.14	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	166.10	3,240,116.00 *gal*
Application event totals		26.19	10.44	215.16	880.24	

Hennes - 11/10/2022: Oats, silage-soft dough

Field name: Hennes

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Hennes - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.62	9.00	83.81	860.54	435,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	287.50	5,448,126.00 *gal*
Application event totals		66.62	9.00	83.81	1,148.04	

02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.69	13.43	276.86	918.93	480,000.00 *gal*
Application event totals		33.69	13.43	276.86	918.93	

03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	18.95	7.55	155.73	516.90	270,000.00 *gal*
Application event totals		18.95	7.55	155.73	516.90	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	26.96	10.74	221.49	735.15	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	249.22	4,722,770.00 *gal*
Application event totals		26.96	10.74	221.49	984.37	

04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.75	15.45	318.39	1,056.77	552,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	217.25	4,117,026.00 *gal*
Application event totals		38.75	15.45	318.39	1,274.03	

Hennes - 06/17/2023: Corn, silage

Field name: Hennes

Crop: Corn, silage

Plant date: 06/17/2023

Hennes - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.69	13.43	276.86	918.93	480,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	281.78	5,339,870.00 *gal*
Application event totals		33.69	13.43	276.86	1,200.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.59	12.59	259.55	861.50	450,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	254.31	4,819,270.00 *gal*
Application event totals		31.59	12.59	259.55	1,115.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.22	17.63	363.38	1,206.10	630,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	234.32	4,440,332.00 *gal*
Application event totals		44.22	17.63	363.38	1,440.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.91	15.11	311.46	1,033.80	540,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.62	4,180,844.00 *gal*
Application event totals		37.91	15.11	311.46	1,254.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.48	11.75	242.25	804.07	420,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	210.99	3,998,332.00 *gal*
Application event totals		29.48	11.75	242.25	1,015.06	

Hennes - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.69	13.43	276.86	918.93	480,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	185.02	3,506,170.00 *gal*
Application event totals		33.69	13.43	276.86	1,103.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.65	20.99	432.59	1,435.83	750,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	168.32	3,189,776.00 *gal*
Application event totals		52.65	20.99	432.59	1,604.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	178.72	3,386,789.00 *gal*
Application event totals		0.00	0.00	0.00	178.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.64	11.42	235.33	781.09	408,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	160.14	3,034,610.00 *gal*
Application event totals		28.64	11.42	235.33	941.23	

Horse Pasture - 11/09/2022: Oats, silage-soft dough

Field name: Horse Pasture

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Horse Pasture - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	242.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	293.73	1,800,854.00 *gal*
Application event totals		0.00	0.00	0.00	293.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	254.43	1,559,888.00 *gal*
Application event totals		0.00	0.00	0.00	254.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	206.79	1,267,849.00 *gal*
Application event totals		0.00	0.00	0.00	206.79	

Horse Pasture - 06/20/2023: Corn, silage

Field name: Horse Pasture

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	308.80	1,893,265.00 *gal*
Application event totals		0.00	0.00	0.00	308.80	

Horse Pasture - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	204.28	73.71	124.25	13,604.76	198.00 *ton*
Application event totals		204.28	73.71	124.25	13,604.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	269.55	1,652,585.00 *gal*
Application event totals		0.00	0.00	0.00	269.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	36.45	14.53	299.51	994.12	168,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	236.65	1,450,878.00 *gal*
Application event totals		36.45	14.53	299.51	1,230.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.05	15.57	320.90	1,065.13	180,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.46	1,351,658.00 *gal*
Application event totals		39.05	15.57	320.90	1,285.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	211.03	1,293,820.00 *gal*
Application event totals		0.00	0.00	0.00	211.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	180.87	1,108,913.00 *gal*
Application event totals		0.00	0.00	0.00	180.87	

Horse Pasture - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	177.78	1,089,961.00 *gal*
Application event totals		0.00	0.00	0.00	177.78	

Manuel N/S - 11/08/2022: Oats, silage-soft dough

Field name: Manuel N/S

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.98	10.13	94.33	968.55	216,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	298.77	2,497,800.00 *gal*
Application event totals		74.98	10.13	94.33	1,267.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.27	6.09	125.51	416.58	96,000.00 *gal*
Application event totals		15.27	6.09	125.51	416.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.71	6.66	137.28	455.64	105,000.00 *gal*
Application event totals		16.71	6.66	137.28	455.64	

Manuel N/S - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.10	15.98	329.46	1,093.53	252,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	266.47	2,227,815.00 *gal*
Application event totals		40.10	15.98	329.46	1,360.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.73	19.03	392.22	1,301.82	300,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	233.62	1,953,150.00 *gal*
Application event totals		47.73	19.03	392.22	1,535.44	

Manuel N/S - 06/18/2023: Corn, silage

Field name: Manuel N/S

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.41	13.32	274.55	911.27	210,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	303.80	2,539,875.00 *gal*
Application event totals		33.41	13.32	274.55	1,215.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	269.52	2,253,300.00 *gal*
Application event totals		0.00	0.00	0.00	269.52	

Manuel N/S - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.51	20.93	431.44	1,432.00	330,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	246.72	2,062,665.00 *gal*
Application event totals		52.51	20.93	431.44	1,678.72	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.96	17.12	352.99	1,171.64	270,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	224.25	1,874,835.00 *gal*
Application event totals		42.96	17.12	352.99	1,395.89	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.80	14.27	294.16	976.37	225,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	213.61	1,785,840.00 *gal*
Application event totals		35.80	14.27	294.16	1,189.97	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.09	7.61	156.89	520.73	120,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	149.05	1,246,113.00 *gal*
Application event totals		19.09	7.61	156.89	669.78	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.73	19.03	392.22	1,301.82	300,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	180.49	1,508,942.00 *gal*
Application event totals		47.73	19.03	392.22	1,482.31	

Manuel N/S - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.91	9.13	188.26	624.87	144,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	131.31	1,097,761.00 *gal*
Application event totals		22.91	9.13	188.26	756.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.19	15.22	313.77	1,041.46	240,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	143.19	1,197,113.00 *gal*
Application event totals		38.19	15.22	313.77	1,184.64	

Manuel S/S - 11/08/2022: Oats, silage-soft dough

Field name: Manuel S/S

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.54	7.50	69.88	717.45	192,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	309.52	3,105,270.00 *gal*
Application event totals		55.54	7.50	69.88	1,026.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.82	12.68	261.48	867.88	240,000.00 *gal*
Application event totals		31.82	12.68	261.48	867.88	

Manuel S/S - 11/08/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	15.91	6.34	130.74	433.94	120,000.00 *gal*
Application event totals		15.91	6.34	130.74	433.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.19	15.22	313.77	1,041.46	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	263.95	2,648,052.00 *gal*
Application event totals		38.19	15.22	313.77	1,305.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.78	15.86	326.85	1,084.85	300,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	221.60	2,223,216.00 *gal*
Application event totals		39.78	15.86	326.85	1,306.45	

Manuel S/S - 06/18/2023: Corn, silage

Field name: Manuel S/S

Crop: Corn, silage

Plant date: 06/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	24.86	9.91	204.28	678.03	187,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	291.35	2,923,002.00 *gal*
Application event totals		24.86	9.91	204.28	969.39	

Manuel S/S - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	267.75	2,686,230.00 *gal*
Application event totals		0.00	0.00	0.00	267.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.78	15.86	326.85	1,084.85	300,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	234.32	2,350,764.00 *gal*
Application event totals		39.78	15.86	326.85	1,319.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.80	14.27	294.16	976.37	270,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.84	2,215,584.00 *gal*
Application event totals		35.80	14.27	294.16	1,197.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.89	7.93	163.42	542.43	150,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	211.36	2,120,490.00 *gal*
Application event totals		19.89	7.93	163.42	753.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.92	20.30	418.36	1,388.61	384,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	187.28	1,878,921.00 *gal*
Application event totals		50.92	20.30	418.36	1,575.89	

Manuel S/S - 06/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.76	17.44	359.53	1,193.34	330,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	164.14	1,646,770.00 *gal*
Application event totals		43.76	17.44	359.53	1,357.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.82	12.68	261.48	867.88	240,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	171.78	1,723,403.00 *gal*
Application event totals		31.82	12.68	261.48	1,039.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.19	15.22	313.77	1,041.46	288,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	150.05	1,505,421.00 *gal*
Application event totals		38.19	15.22	313.77	1,191.51	

Marks Trees - 01/01/2023: Almonds

Field name: Marks Trees

Crop: Almonds

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	361.41	2,215,800.00 *gal*
Application event totals		0.00	0.00	0.00	361.41	

Marks Trees - 01/01/2023: Almonds

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Can 17	Solid commercial fertilizer	68.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	314.65	1,929,110.00 *gal*
Application event totals		68.00	0.00	0.00	314.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	242.89	1,489,120.00 *gal*
Application event totals		0.00	0.00	0.00	242.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
Canal	Surface water	0.00	0.00	0.00	219.52	1,345,870.00 *gal*
Application event totals		80.00	0.00	0.00	219.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	196.55	1,205,022.00 *gal*
Application event totals		0.00	0.00	0.00	196.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	253.14	1,552,001.00 *gal*
Application event totals		0.00	0.00	0.00	253.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	192.38	1,179,500.00 *gal*
Application event totals		0.00	0.00	0.00	192.38	

Marks Trees - 01/01/2023: Almonds

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	197.39	1,210,200.00 *gal*
Application event totals		0.00	0.00	0.00	197.39	

Morris - 11/09/2022: Oats, silage-soft dough

Field name: Morris

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	396.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	284.76	2,856,870.00 *gal*
Application event totals		0.00	0.00	0.00	284.76	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	227.50	2,282,418.00 *gal*
Application event totals		0.00	0.00	0.00	227.50	

Morris - 06/17/2023: Corn, silage

Field name: Morris

Crop: Corn, silage

Plant date: 06/17/2023

Morris - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	204.28	73.71	124.25	13,604.76	324.00 *ton*
Application event totals		204.28	73.71	124.25	13,604.76	

06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.83	11.89	245.13	813.64	225,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	299.55	3,005,244.00 *gal*
Application event totals		29.83	11.89	245.13	1,113.19	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	272.93	2,738,160.00 *gal*
Application event totals		0.00	0.00	0.00	272.93	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	250.57	2,513,844.00 *gal*
Application event totals		0.00	0.00	0.00	250.57	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.72	20.22	416.73	1,383.18	382,500.00 *gal*
Canal	Surface water	0.00	0.00	0.00	229.81	2,305,584.00 *gal*
Application event totals		50.72	20.22	416.73	1,613.00	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	220.51	2,212,290.00 *gal*
Application event totals		0.00	0.00	0.00	220.51	

Morris - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	218.21	2,189,216.00 *gal*
Application event totals		0.00	0.00	0.00	218.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	219.50	2,202,140.00 *gal*
Application event totals		0.00	0.00	0.00	219.50	

Pasture - 11/09/2022: Oats, silage-soft dough

Field name: Pasture

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	748.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	308.63	5,848,510.00 *gal*
Application event totals		0.00	0.00	0.00	308.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	254.97	4,831,672.00 *gal*
Application event totals		0.00	0.00	0.00	254.97	

Pasture - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	220.90	4,186,026.00 *gal*
Application event totals		0.00	0.00	0.00	220.90	

Pasture - 06/17/2023: Corn, silage

Field name: Pasture

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	204.28	73.71	124.25	13,604.76	612.00 *ton*
Application event totals		204.28	73.71	124.25	13,604.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.12	16.79	346.07	1,148.66	600,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	291.28	5,519,900.00 *gal*
Application event totals		42.12	16.79	346.07	1,439.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	269.55	5,107,990.00 *gal*
Application event totals		0.00	0.00	0.00	269.55	

Pasture - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.54	13.77	283.78	941.91	492,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	250.95	4,755,512.00 *gal*
Application event totals		34.54	13.77	283.78	1,192.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	232.61	4,408,100.00 *gal*
Application event totals		0.00	0.00	0.00	232.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	215.52	4,084,080.00 *gal*
Application event totals		0.00	0.00	0.00	215.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	204.33	3,872,113.00 *gal*
Application event totals		0.00	0.00	0.00	204.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	195.91	3,712,613.00 *gal*
Application event totals		0.00	0.00	0.00	195.91	

Silveria 1 - 11/09/2022: Oats, silage-soft dough

Field name: Silveria 1

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Silveria 1 - 11/09/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	110.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	287.07	800,000.00 *gal*
Application event totals		0.00	0.00	0.00	287.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.70	779,475.00 *gal*
Application event totals		0.00	0.00	0.00	279.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	234.50	653,490.00 *gal*
Application event totals		0.00	0.00	0.00	234.50	

Silveria 1 - 06/20/2023: Corn, silage

Field name: Silveria 1

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	204.28	73.71	124.25	13,604.76	90.00 *ton*
Application event totals		204.28	73.71	124.25	13,604.76	

Silveria 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.37	13.70	282.39	937.31	72,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	294.45	820,575.00 *gal*
Application event totals		34.37	13.70	282.39	1,231.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	272.63	759,755.00 *gal*
Application event totals		0.00	0.00	0.00	272.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	11.46	4.57	94.13	312.44	24,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	240.81	671,080.00 *gal*
Application event totals		11.46	4.57	94.13	553.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	221.96	618,560.00 *gal*
Application event totals		0.00	0.00	0.00	221.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	217.27	605,495.00 *gal*
Application event totals		0.00	0.00	0.00	217.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	215.06	599,340.00 *gal*
Application event totals		0.00	0.00	0.00	215.06	

Silveria 1 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.91	9.13	188.26	624.87	48,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.46	600,445.00 *gal*
Application event totals		22.91	9.13	188.26	840.33	

Silveria 2 - 11/09/2022: Oats, silage-soft dough

Field name: Silveria 2

Crop: Oats, silage-soft dough

Plant date: 11/09/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	176.18	36.78	41.62	3,223.44	506.00 *ton*
Application event totals		176.18	36.78	41.62	3,223.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	287.07	3,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	287.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	279.70	3,585,585.00 *gal*
Application event totals		0.00	0.00	0.00	279.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	234.50	3,006,054.00 *gal*
Application event totals		0.00	0.00	0.00	234.50	

Silveria 2 - 06/20/2023: Corn, silage

Field name: Silveria 2

Crop: Corn, silage

Plant date: 06/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	204.28	73.71	124.25	13,604.76	414.00 *ton*
Application event totals		204.28	73.71	124.25	13,604.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.38	12.90	266.02	882.97	312,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	294.45	3,774,645.00 *gal*
Application event totals		32.38	12.90	266.02	1,177.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	272.63	3,494,873.00 *gal*
Application event totals		0.00	0.00	0.00	272.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	16.19	6.45	133.01	441.49	156,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	240.81	3,086,968.00 *gal*
Application event totals		16.19	6.45	133.01	682.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	221.96	2,845,376.00 *gal*
Application event totals		0.00	0.00	0.00	221.96	

Silveria 2 - 06/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	217.27	2,785,277.00 *gal*
Application event totals		0.00	0.00	0.00	217.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	215.06	2,756,964.00 *gal*
Application event totals		0.00	0.00	0.00	215.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	21.17	8.44	173.94	577.33	204,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	215.46	2,762,047.00 *gal*
Application event totals		21.17	8.44	173.94	792.79	

The 60 - 11/07/2022: Oats, silage-soft dough

Field name: The 60

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	278.91	10,104,510.00 *gal*
Application event totals		0.00	0.00	0.00	278.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.13	12.99	120.94	1,241.74	1,200,000.00 *gal*
Application event totals		96.13	12.99	120.94	1,241.74	

The 60 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	17.07	6.81	140.29	465.65	465,000.00 *gal*
Application event totals		17.07	6.81	140.29	465.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.91	9.13	188.26	624.87	624,000.00 *gal*
Application event totals		22.91	9.13	188.26	624.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.35	19.67	405.49	1,345.88	1,344,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	239.09	8,661,835.00 *gal*
Application event totals		49.35	19.67	405.49	1,584.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.39	7.73	159.30	528.74	528,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	220.07	7,972,640.00 *gal*
Application event totals		19.39	7.73	159.30	748.81	

The 60 - 06/19/2023: Corn, silage

Field name: The 60

Crop: Corn, silage

Plant date: 06/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

The 60 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported SM	Separator solids	129.95	59.98	35.87	1,099.56	1,300.00 *ton*
Application event totals		129.95	59.98	35.87	1,099.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.30	16.86	347.56	1,153.61	1,152,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	305.40	11,063,975.00 *gal*
Application event totals		42.30	16.86	347.56	1,459.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	274.69	9,951,435.00 *gal*
Application event totals		0.00	0.00	0.00	274.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.66	15.81	325.84	1,081.51	1,080,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	239.75	8,685,690.00 *gal*
Application event totals		39.66	15.81	325.84	1,321.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	218.49	7,915,375.00 *gal*
Application event totals		0.00	0.00	0.00	218.49	

The 60 - 06/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.37	18.88	389.20	1,291.81	1,290,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	198.78	7,201,321.00 *gal*
Application event totals		47.37	18.88	389.20	1,490.58	

08/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	187.94	6,808,917.00 *gal*
Application event totals		0.00	0.00	0.00	187.94	

09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	182.92	6,626,921.00 *gal*
Application event totals		0.00	0.00	0.00	182.92	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.77	15.46	318.60	1,057.48	1,056,000.00 *gal*
Canal	Surface water	0.00	0.00	0.00	157.53	5,707,114.00 *gal*
Application event totals		38.77	15.46	318.60	1,215.01	

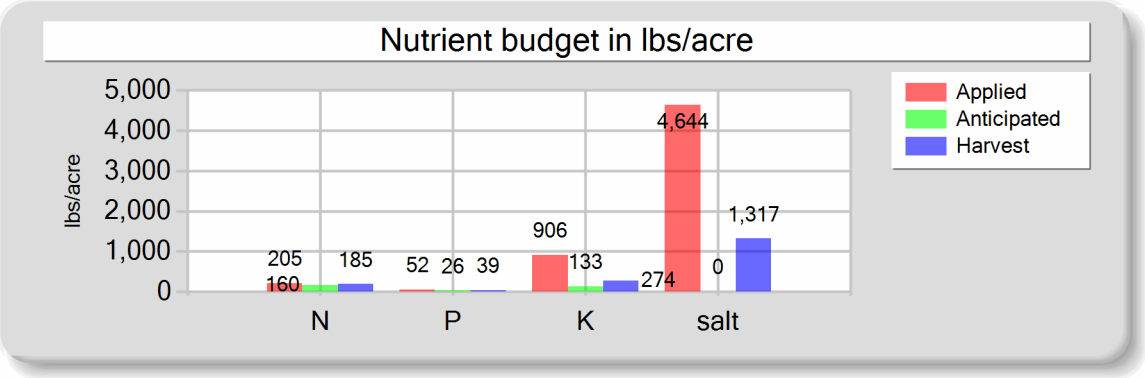
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.00	0.00	0.00	151.83	5,500,721.00 *gal*
Application event totals		0.00	0.00	0.00	151.83	

B. NUTRIENT BUDGET

50 - 11/10/2022: Oats, silage-soft dough

Field name: 50 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	197.62	51.60	906.29	3,909.37
Fresh water	0.00	0.00	0.00	734.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	204.62	51.60	906.29	4,643.74
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	185.18	39.17	274.21	1,317.34
Nutrient balance	19.44	12.43	632.08	3,326.39
Applied to removed ratio	1.10	1.32	3.31	3.53

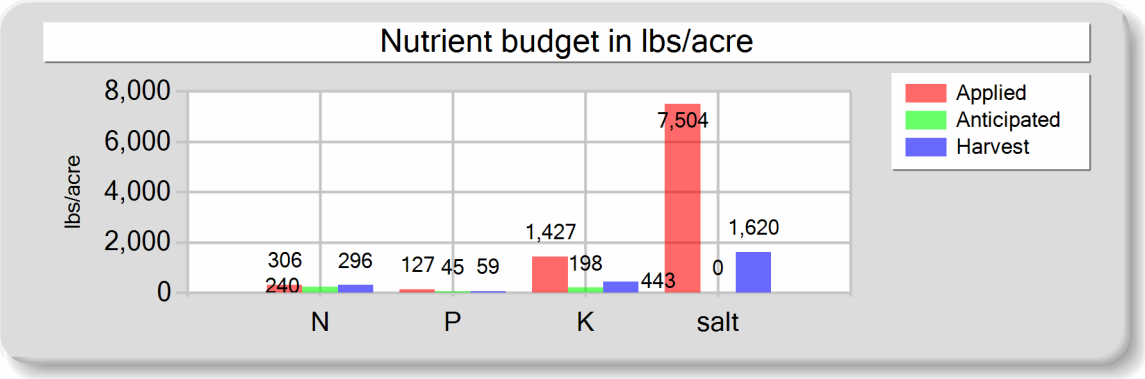
Fresh water applied	
20,465,450.00	*gallons*
753.67	*acre-inches*
15.07	*inches/acre*

Process wastewater applied	
2,970,000.00	*gallons*
109.38	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

50 - 06/19/2023: Corn, silage

Field name: 50 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.95	59.98	35.87	1,099.56
Process wastewater	169.26	67.47	1,390.79	4,616.25
Fresh water	0.00	0.00	0.00	1,787.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.21	127.44	1,426.66	7,503.70
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	295.59	59.12	443.39	1,620.08
Nutrient balance	10.62	68.33	983.27	5,883.62
Applied to removed ratio	1.04	2.16	3.22	4.63

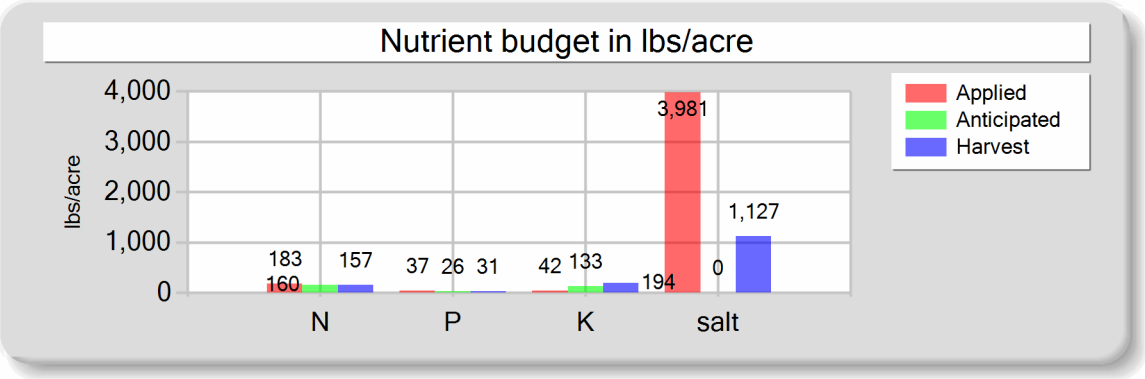
Fresh water applied	
49,824,832.00	*gallons*
1,834.88	*acre-inches*
36.70	*inches/acre*

Process wastewater applied	
3,546,000.00	*gallons*
130.59	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

70 - 11/07/2022: Oats, silage-soft dough

Field name: 70 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	757.36
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	3,980.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.31	30.90	193.83	1,126.87
Nutrient balance	25.87	5.88	-152.20	2,853.93
Applied to removed ratio	1.16	1.19	0.21	3.53

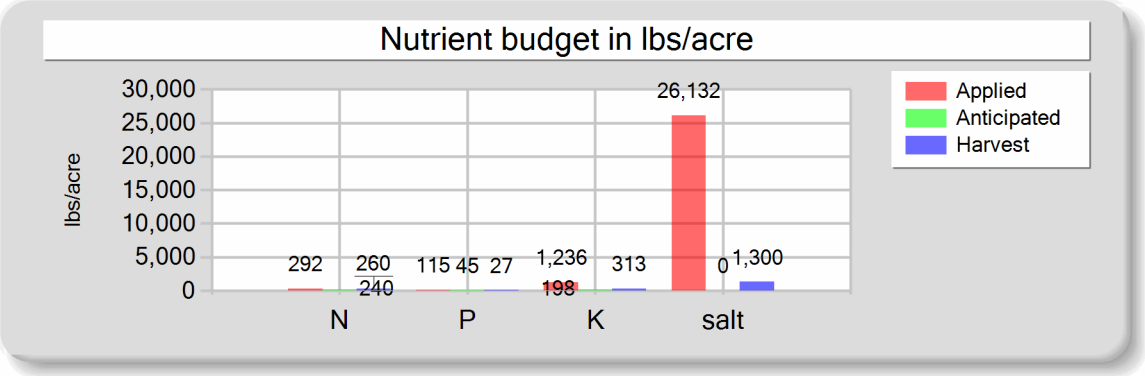
Fresh water applied	
27,437,930.00	*gallons*
1,010.45	*acre-inches*
15.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

70 - 06/20/2023: Corn, silage

Field name: 70 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	149.54	60.92	124.62	20,796.92
Process wastewater	135.27	53.92	1,111.48	3,689.16
Fresh water	0.00	0.00	0.00	1,646.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.81	114.84	1,236.09	26,132.43
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	259.93	26.52	312.97	1,299.63
Nutrient balance	31.88	88.32	923.12	24,832.80
Applied to removed ratio	1.12	4.33	3.95	20.11

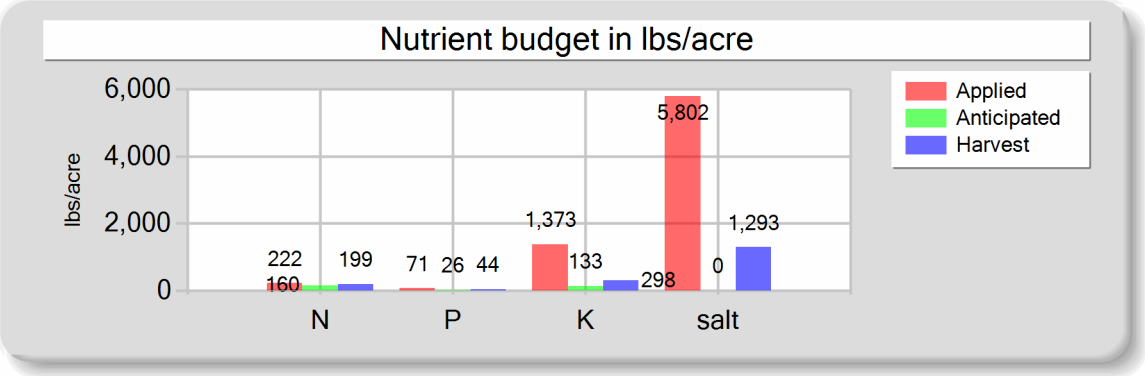
Fresh water applied	
59,644,630.00	*gallons*
2,196.51	*acre-inches*
33.79	*inches/acre*

Process wastewater applied	
3,684,000.00	*gallons*
135.67	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

Corn Shed - 11/08/2022: Oats, silage-soft dough

Field name: Corn Shed | Crop: Oats, silage-soft dough | Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.47	70.83	1,372.91	5,056.27
Fresh water	0.00	0.00	0.00	745.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	222.47	70.83	1,372.91	5,801.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	198.62	44.14	297.93	1,293.22
Nutrient balance	23.85	26.69	1,074.98	4,508.35
Applied to removed ratio	1.12	1.60	4.61	4.49

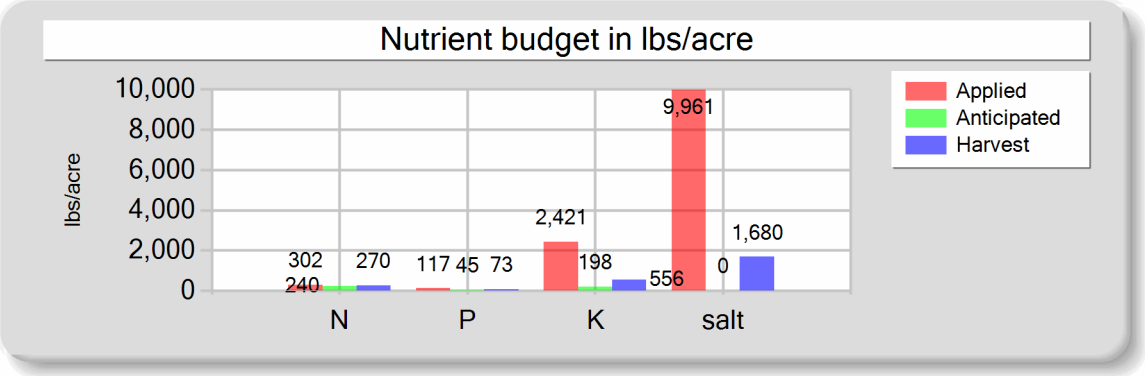
Fresh water applied	
14,539,035.00	*gallons*
535.42	*acre-inches*
15.30	*inches/acre*

Process wastewater applied	
2,706,000.00	*gallons*
99.65	*acre-inches*
2.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Corn Shed - 06/17/2023: Corn, silage

Field name: Corn Shed Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	294.58	117.42	2,420.53	8,034.09
Fresh water	0.00	0.00	0.00	1,927.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	301.58	117.42	2,420.53	9,961.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	270.40	72.80	556.41	1,679.62
Nutrient balance	31.18	44.62	1,864.12	8,281.59
Applied to removed ratio	1.12	1.61	4.35	5.93

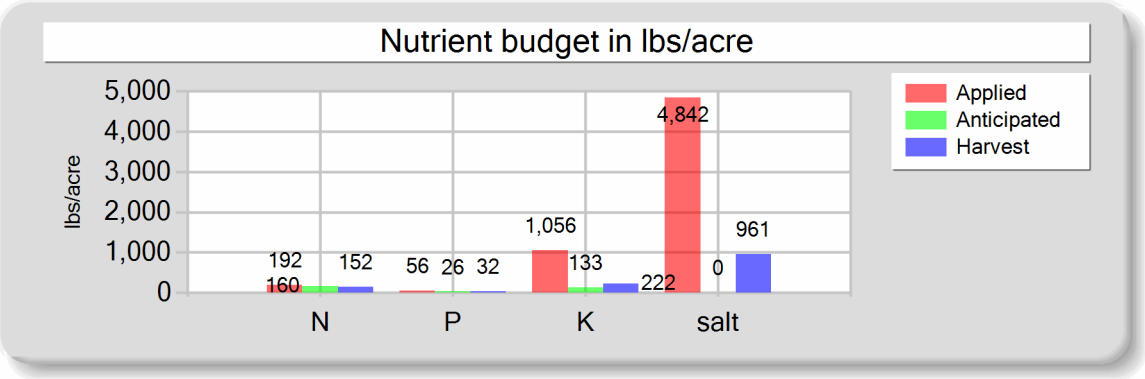
Fresh water applied	
37,593,461.00	*gallons*
1,384.44	*acre-inches*
39.56	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
4.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hennes - 11/10/2022: Oats, silage-soft dough

Field name: Hennes | Crop: Oats, silage-soft dough | Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.97	56.18	1,056.27	4,088.29
Fresh water	0.00	0.00	0.00	753.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.97	56.18	1,056.27	4,842.26
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	151.68	32.09	221.69	961.02
Nutrient balance	40.29	24.09	834.59	3,881.25
Applied to removed ratio	1.27	1.75	4.76	5.04

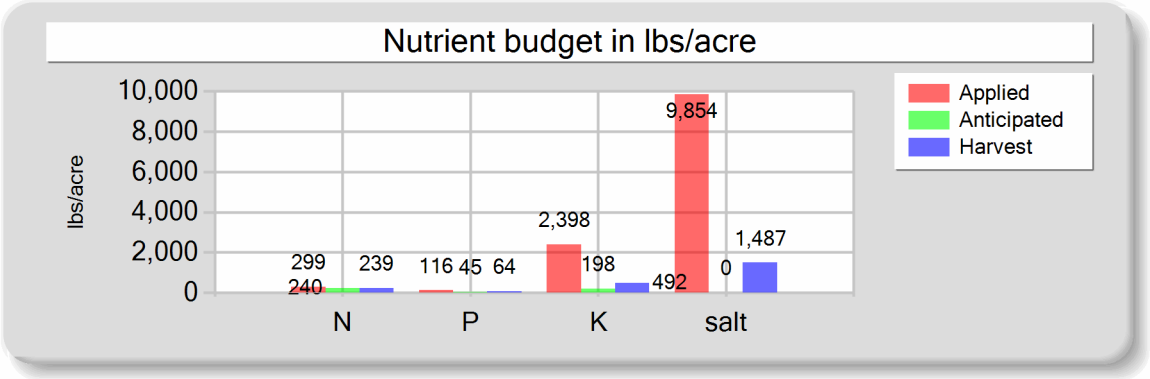
Fresh water applied	
14,287,922.00	*gallons*
526.18	*acre-inches*
15.48	*inches/acre*

Process wastewater applied	
2,121,000.00	*gallons*
78.11	*acre-inches*
2.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Hennes - 06/17/2023: Corn, silage

Field name: Hennes | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	291.88	116.34	2,398.28	7,960.25
Fresh water	0.00	0.00	0.00	1,894.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	298.88	116.34	2,398.28	9,854.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.32	64.43	492.45	1,486.54
Nutrient balance	59.56	51.91	1,905.83	8,367.93
Applied to removed ratio	1.25	1.81	4.87	6.63

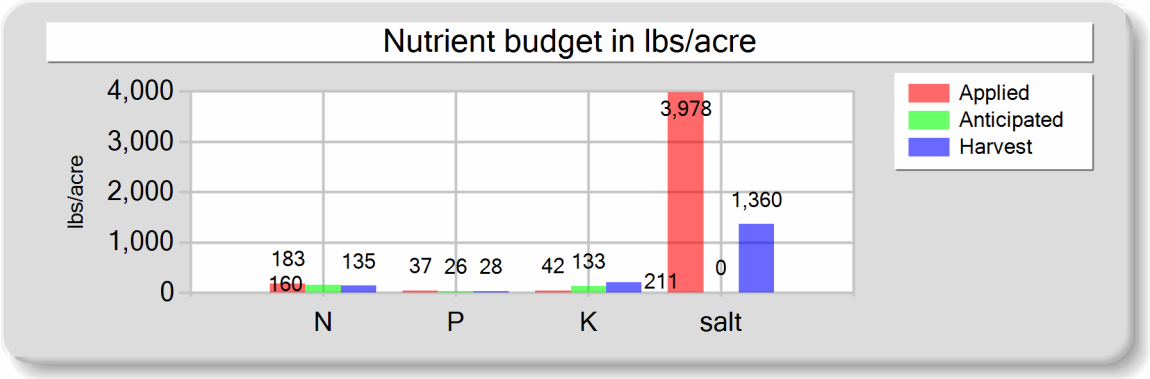
Fresh water applied	
35,895,993.00	*gallons*
1,321.93	*acre-inches*
38.88	*inches/acre*

Process wastewater applied	
4,158,000.00	*gallons*
153.13	*acre-inches*
4.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

Horse Pasture - 11/09/2022: Oats, silage-soft dough

Field name: Horse Pasture Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	754.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	3,978.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.74	27.64	210.75	1,359.85
Nutrient balance	48.43	9.14	-169.13	2,618.54
Applied to removed ratio	1.36	1.33	0.20	2.93

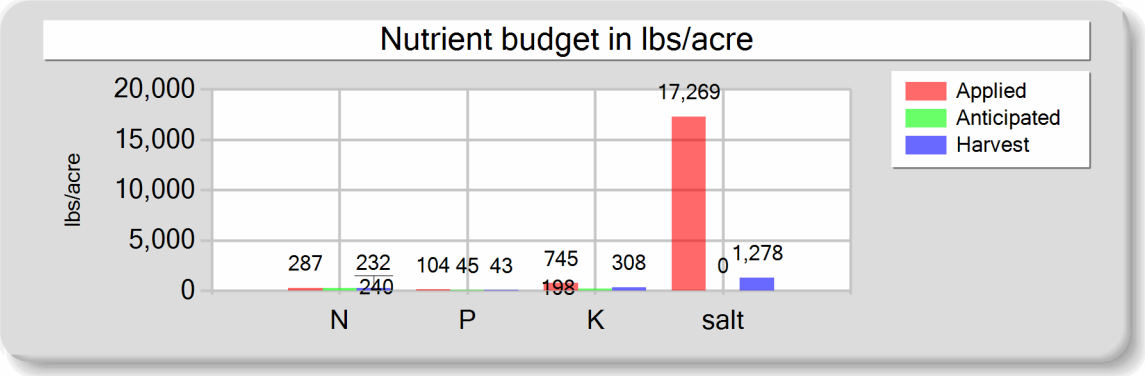
Fresh water applied	
4,628,591.00	*gallons*
170.46	*acre-inches*
15.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Horse Pasture - 06/20/2023: Corn, silage

Field name: Horse Pasture Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.28	73.71	124.25	13,604.76
Process wastewater	75.51	30.10	620.41	2,059.24
Fresh water	0.00	0.00	0.00	1,605.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	286.79	103.81	744.67	17,269.15
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	232.25	43.21	307.87	1,277.89
Nutrient balance	54.53	60.60	436.79	15,991.26
Applied to removed ratio	1.23	2.40	2.42	13.51

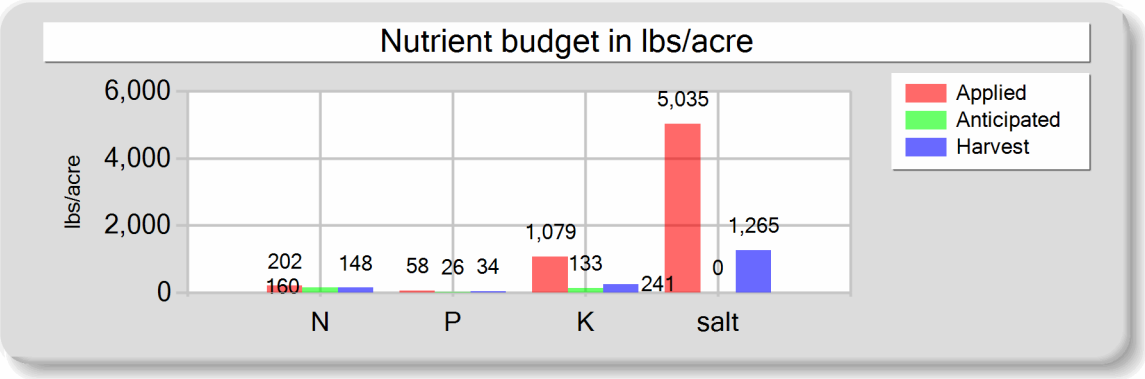
Fresh water applied	
9,841,080.00	*gallons*
362.41	*acre-inches*
32.95	*inches/acre*

Process wastewater applied	
348,000.00	*gallons*
12.82	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manuel N/S - 11/08/2022: Oats, silage-soft dough

Field name: Manuel N/S Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.80	57.89	1,078.79	4,236.12
Fresh water	0.00	0.00	0.00	798.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.80	57.89	1,078.79	5,034.98
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.03	34.43	240.99	1,264.84
Nutrient balance	53.76	23.46	837.80	3,770.14
Applied to removed ratio	1.36	1.68	4.48	3.98

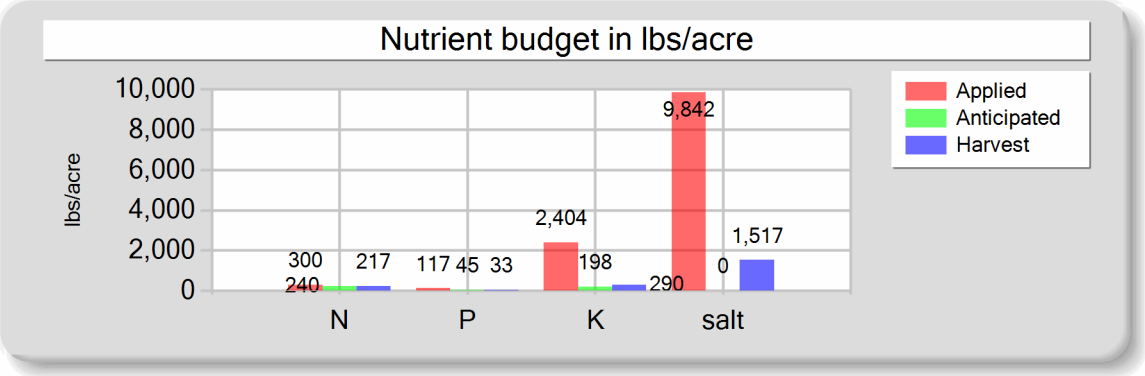
Fresh water applied	
6,678,765.00	*gallons*
245.96	*acre-inches*
16.40	*inches/acre*

Process wastewater applied	
969,000.00	*gallons*
35.68	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manuel N/S - 06/18/2023: Corn, silage

Field name: Manuel N/S Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	292.61	116.63	2,404.28	7,980.16
Fresh water	0.00	0.00	0.00	1,861.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	299.61	116.63	2,404.28	9,842.08
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	217.32	33.43	289.76	1,517.09
Nutrient balance	82.28	83.20	2,114.51	8,325.00
Applied to removed ratio	1.38	3.49	8.30	6.49

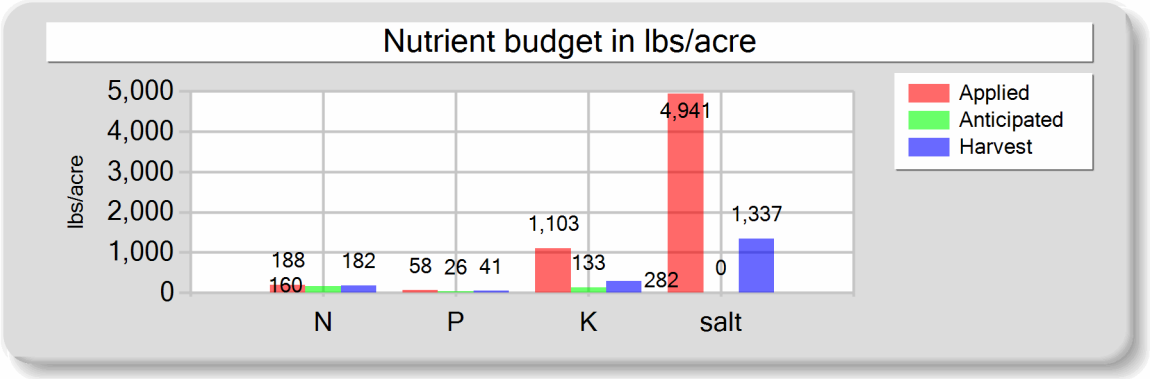
Fresh water applied	
15,566,444.00	*gallons*
573.26	*acre-inches*
38.22	*inches/acre*

Process wastewater applied	
1,839,000.00	*gallons*
67.72	*acre-inches*
4.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manuel S/S - 11/08/2022: Oats, silage-soft dough

Field name: Manuel S/S Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.24	57.61	1,102.71	4,145.57
Fresh water	0.00	0.00	0.00	795.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.24	57.61	1,102.71	4,940.65
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	182.46	41.31	282.30	1,336.79
Nutrient balance	5.78	16.30	820.41	3,603.86
Applied to removed ratio	1.03	1.39	3.91	3.70

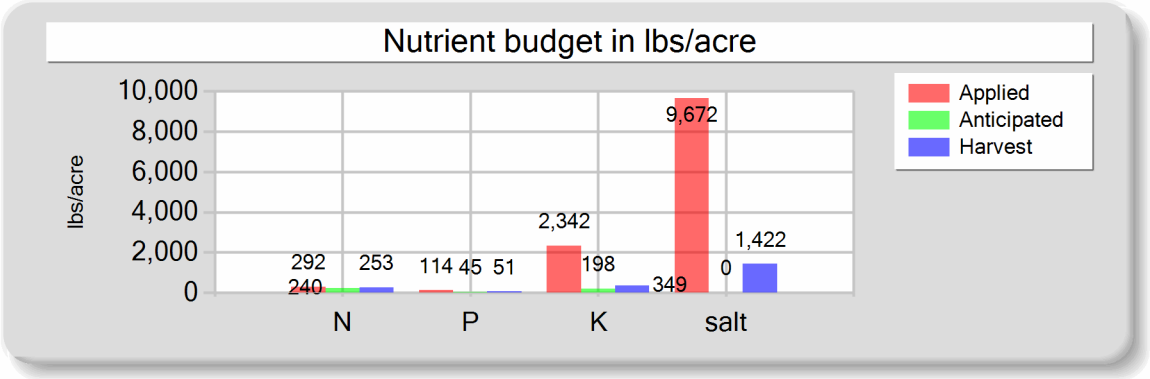
Fresh water applied	
7,976,538.00	*gallons*
293.75	*acre-inches*
16.32	*inches/acre*

Process wastewater applied	
1,140,000.00	*gallons*
41.98	*acre-inches*
2.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Manuel S/S - 06/18/2023: Corn, silage

Field name: Manuel S/S Crop: Corn, silage Plant date: 06/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.01	113.60	2,341.85	7,772.95
Fresh water	0.00	0.00	0.00	1,898.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.01	113.60	2,341.85	9,671.84
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.35	50.67	349.05	1,422.28
Nutrient balance	38.66	62.94	1,992.80	8,249.56
Applied to removed ratio	1.15	2.24	6.71	6.80

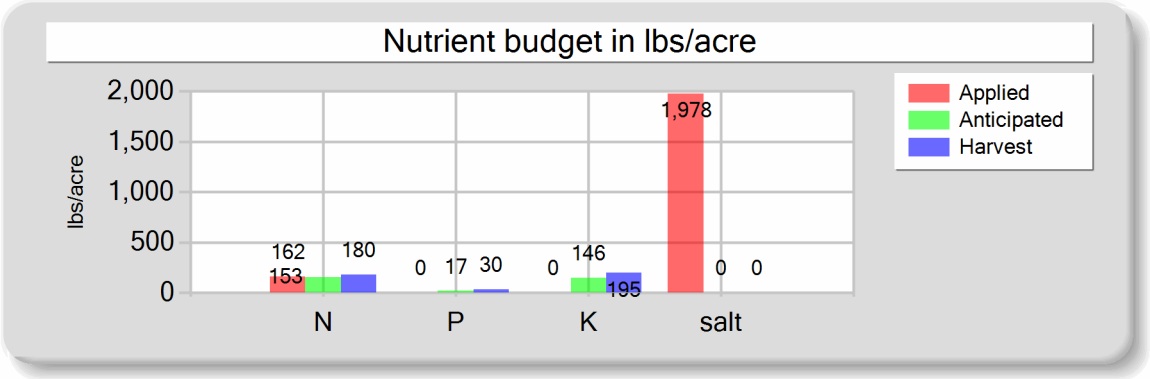
Fresh water applied	
19,050,585.00	*gallons*
701.57	*acre-inches*
38.98	*inches/acre*

Process wastewater applied	
2,149,500.00	*gallons*
79.16	*acre-inches*
4.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Marks Trees - 01/01/2023: Almonds

Field name: Marks Trees Crop: Almonds Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	148.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,977.93
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	162.00	0.00	0.00	1,977.93
Anticipated crop nutrient removal	152.80	16.80	146.40	0.00
Actual crop nutrient removal	179.64	30.40	194.84	0.00
Nutrient balance	-17.64	-30.40	-194.84	1,977.93
Applied to removed ratio	0.90	0.00	0.00	0.00

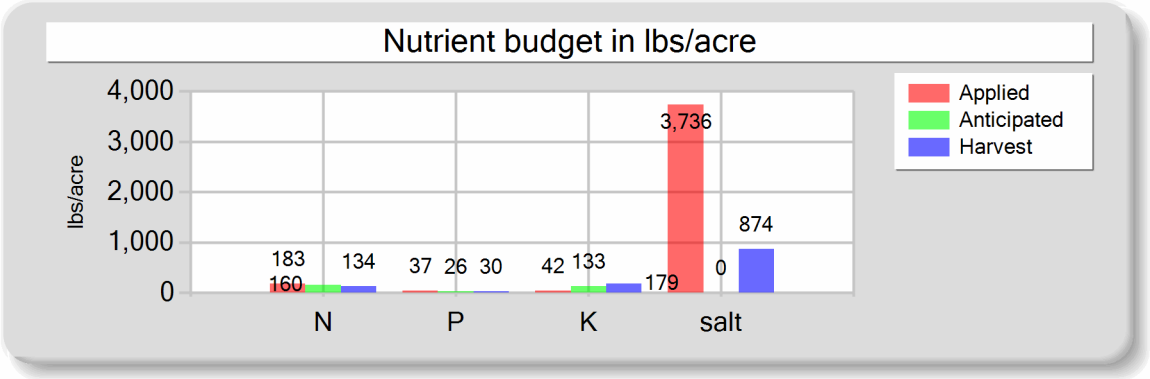
Fresh water applied	
12,126,623.00	*gallons*
446.58	*acre-inches*
40.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 11/09/2022: Oats, silage-soft dough

Field name: Morris | Crop: Oats, silage-soft dough | Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	512.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	3,735.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.35	29.85	179.13	873.65
Nutrient balance	48.83	6.93	-137.50	2,862.05
Applied to removed ratio	1.36	1.23	0.23	4.28

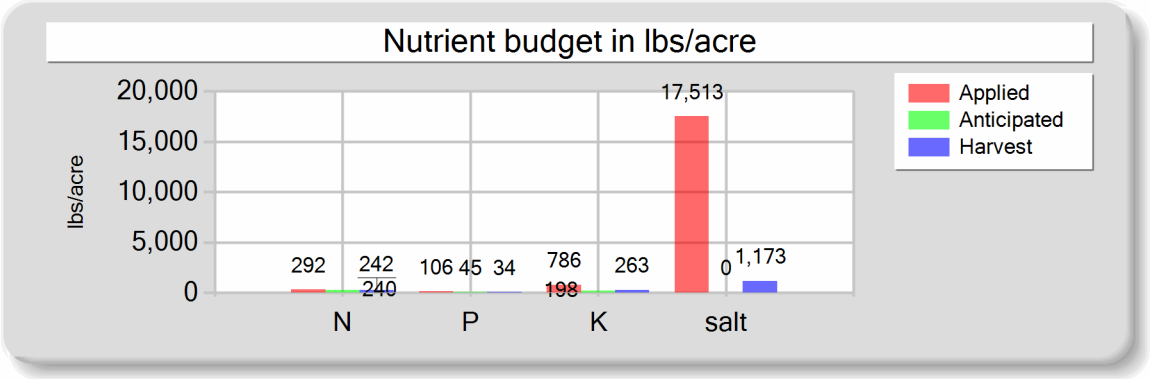
Fresh water applied	
5,139,288.00	*gallons*
189.26	*acre-inches*
10.51	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 06/17/2023: Corn, silage

Field name: Morris | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.28	73.71	124.25	13,604.76
Process wastewater	80.55	32.11	661.86	2,196.82
Fresh water	0.00	0.00	0.00	1,711.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.83	105.82	786.12	17,512.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.73	33.93	262.94	1,173.50
Nutrient balance	50.10	71.89	523.18	16,339.18
Applied to removed ratio	1.21	3.12	2.99	14.92

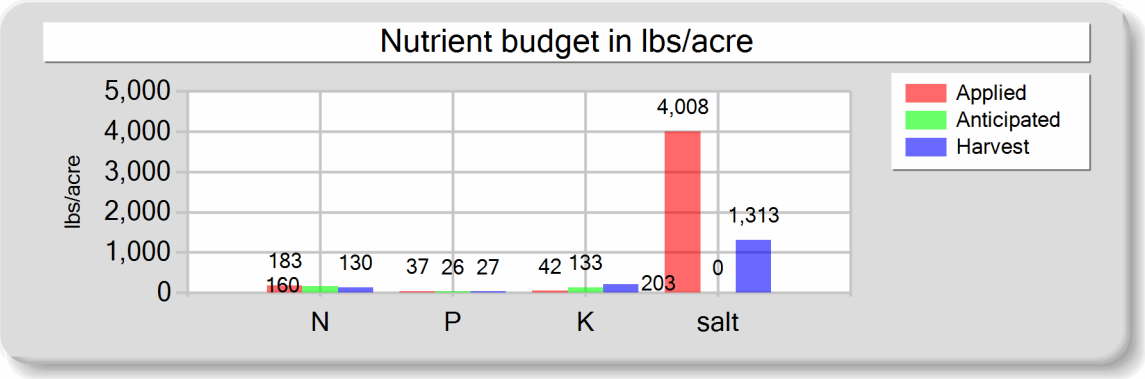
Fresh water applied	
17,166,478.00	*gallons*
632.18	*acre-inches*
35.12	*inches/acre*

Process wastewater applied	
607,500.00	*gallons*
22.37	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pasture - 11/09/2022: Oats, silage-soft dough

Field name: Pasture | Crop: Oats, silage-soft dough | Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	784.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	4,007.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	130.06	26.68	203.43	1,312.63
Nutrient balance	53.11	10.10	-161.81	2,695.29
Applied to removed ratio	1.41	1.38	0.20	3.05

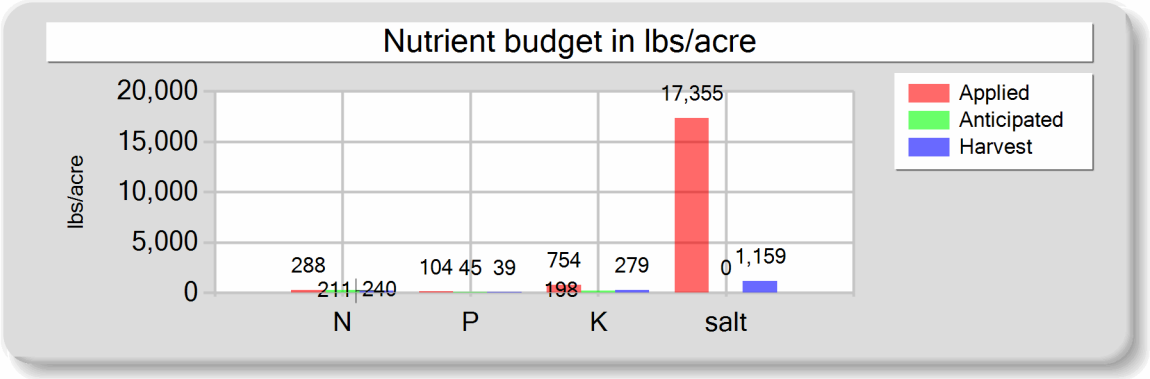
Fresh water applied	
14,866,208.00	*gallons*
547.47	*acre-inches*
16.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Pasture - 06/17/2023: Corn, silage

Field name: Pasture | Crop: Corn, silage | Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.28	73.71	124.25	13,604.76
Process wastewater	76.65	30.55	629.85	2,090.57
Fresh water	0.00	0.00	0.00	1,660.16
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	287.94	104.26	754.11	17,355.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	210.70	39.20	279.30	1,159.29
Nutrient balance	77.24	65.06	474.81	16,196.19
Applied to removed ratio	1.37	2.66	2.70	14.97

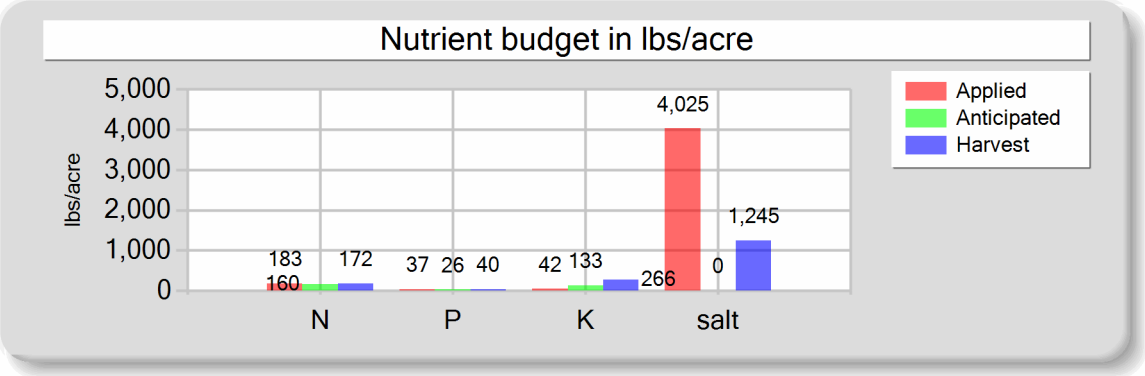
Fresh water applied	
31,460,308.00	*gallons*
1,158.58	*acre-inches*
34.08	*inches/acre*

Process wastewater applied	
1,092,000.00	*gallons*
40.21	*acre-inches*
1.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silveria 1 - 11/09/2022: Oats, silage-soft dough

Field name: Silveria 1 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	801.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	4,024.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	171.89	40.44	266.26	1,245.40
Nutrient balance	11.29	-3.66	-224.64	2,779.31
Applied to removed ratio	1.07	0.91	0.16	3.23

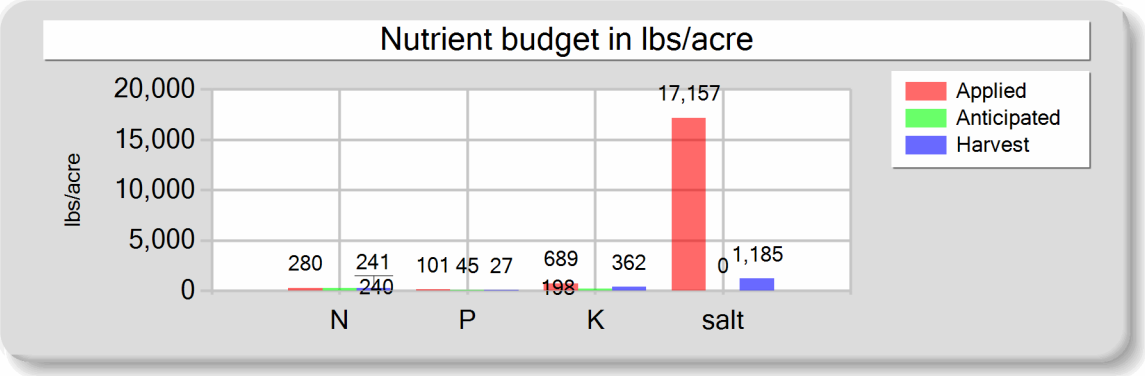
Fresh water applied
2,232,965.00 *gallons*
82.23 *acre-inches*
16.45 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Silveria 1 - 06/20/2023: Corn, silage

Field name: Silveria 1 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.28	73.71	124.25	13,604.76
Process wastewater	68.74	27.40	564.79	1,874.62
Fresh water	0.00	0.00	0.00	1,677.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.02	101.11	689.04	17,157.02
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	241.12	27.40	361.68	1,184.50
Nutrient balance	38.90	73.71	327.36	15,972.52
Applied to removed ratio	1.16	3.69	1.91	14.48

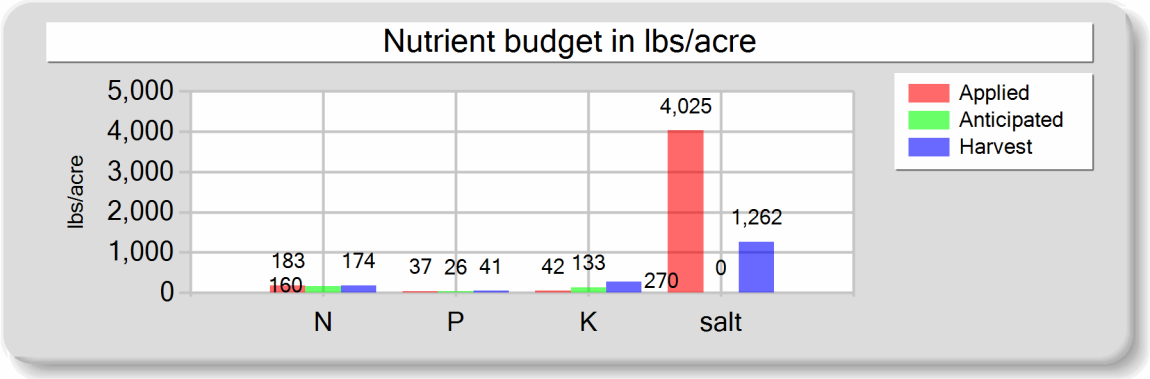
Fresh water applied	
4,675,250.00	*gallons*
172.17	*acre-inches*
34.43	*inches/acre*

Process wastewater applied	
144,000.00	*gallons*
5.30	*acre-inches*
1.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silveria 2 - 11/09/2022: Oats, silage-soft dough

Field name: Silveria 2 Crop: Oats, silage-soft dough Plant date: 11/09/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	176.18	36.78	41.62	3,223.44
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	801.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.18	36.78	41.62	4,024.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	174.17	40.98	269.79	1,261.89
Nutrient balance	9.01	-4.20	-228.16	2,762.81
Applied to removed ratio	1.05	0.90	0.15	3.19

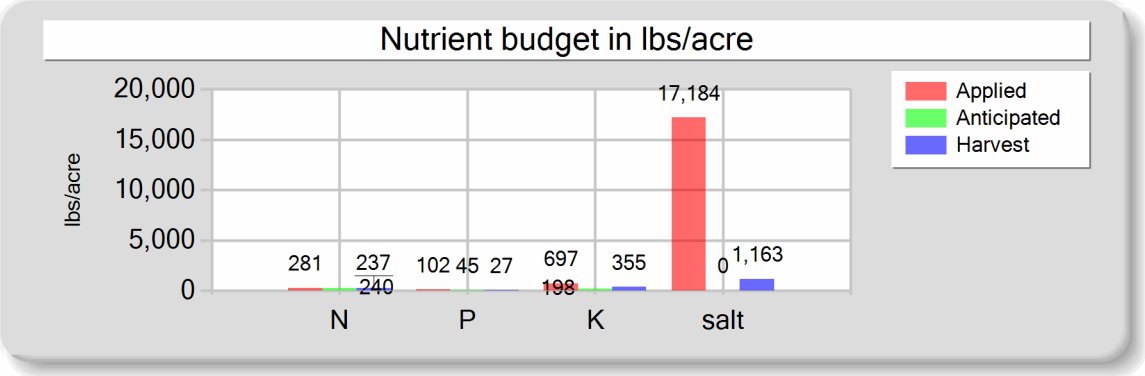
Fresh water applied	
10,271,639.00	*gallons*
378.27	*acre-inches*
16.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Silveria 2 - 06/20/2023: Corn, silage

Field name: Silveria 2 Crop: Corn, silage Plant date: 06/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	204.28	73.71	124.25	13,604.76
Process wastewater	69.73	27.80	572.97	1,901.79
Fresh water	0.00	0.00	0.00	1,677.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.01	101.51	697.23	17,184.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.83	26.91	355.25	1,163.45
Nutrient balance	44.18	74.59	341.98	16,020.74
Applied to removed ratio	1.19	3.77	1.96	14.77

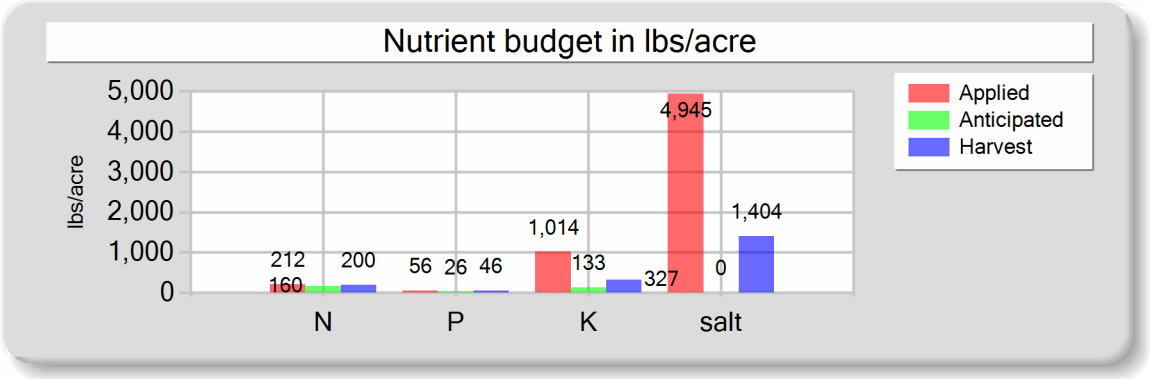
Fresh water applied	
21,506,150.00	*gallons*
792.00	*acre-inches*
34.43	*inches/acre*

Process wastewater applied	
672,000.00	*gallons*
24.75	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 60 - 11/07/2022: Oats, silage-soft dough

Field name: The 60 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	204.86	56.32	1,014.28	4,206.88
Fresh water	0.00	0.00	0.00	738.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.86	56.32	1,014.28	4,944.95
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	199.96	46.14	326.86	1,404.33
Nutrient balance	11.90	10.18	687.43	3,540.61
Applied to removed ratio	1.06	1.22	3.10	3.52

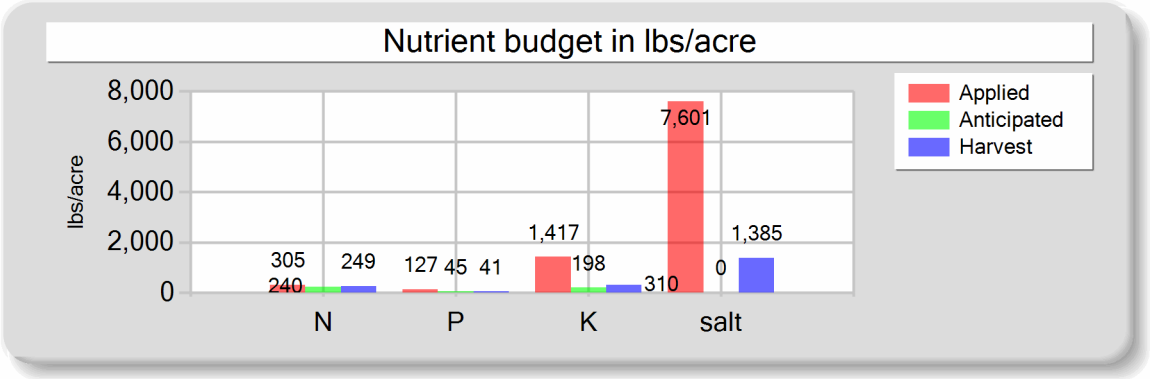
Fresh water applied	
26,738,985.00	*gallons*
984.71	*acre-inches*
15.15	*inches/acre*

Process wastewater applied	
4,161,000.00	*gallons*
153.24	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 60 - 06/19/2023: Corn, silage

Field name: The 60 Crop: Corn, silage Plant date: 06/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.95	59.98	35.87	1,099.56
Process wastewater	168.10	67.00	1,381.20	4,584.41
Fresh water	0.00	0.00	0.00	1,917.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.04	126.98	1,417.07	7,601.29
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.36	40.71	310.43	1,385.02
Nutrient balance	55.68	86.27	1,106.64	6,216.27
Applied to removed ratio	1.22	3.12	4.56	5.49

Fresh water applied	
69,461,469.00	*gallons*
2,558.03	*acre-inches*
39.35	*inches/acre*

Process wastewater applied	
4,578,000.00	*gallons*
168.59	*acre-inches*
2.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Corral Solids

Sample and source description: 2022 Fall Corral Solids

Sample date: 11/15/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	4,800.00	6,500.00							14.70
DL	100.00	100.00	100.00							0.01

2022 Fall Separated Solids

Sample and source description: 2022 Fall Separated Solids

Sample date: 11/15/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,800.00	4,300.00	15,800.00	4,700.00	1,800.00	3,700.00	1,010.00		33.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

Imported Solid Manure CMC Jerseys

Sample and source description: Imported Solid Manure CMC Jerseys

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 85.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	10,200.00	6,100.00							18.70
DL	100.00	100.00	100.00							0.01

Imported Solid Manure DC #6

Sample and source description: Imported Solid Manure DC #6

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 55.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	2,200.00	4,500.00							75.10
DL	100.00	100.00	100.00							0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids

Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,700.00	3,500.00	5,900.00							64.60
DL	100.00	100.00	100.00							0.01

Spring Separated Solids

Sample and source description: Spring Separated Solids

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,500.00	1,800.00	2,400.00							71.40
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids

Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	4,200.00	7,800.00							23.80
DL	100.00	100.00	100.00							0.01

Fall Separated Solids

Sample and source description: Fall Separated Solids

Sample date: 10/23/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	3,000.00	5,100.00	9,600.00	3,700.00	2,100.00	2,900.00	673.00		13.50
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/12/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	624.00	468.00	0.00	0.00	84.30	785.00								7,270.00	8,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,040.00	423.00	0.00	0.00	78.20	655.00								6,120.00	5,320
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	286.00	84.00	0.00	0.00	114.00	2,350.00								8,720.00	7,800
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	143.00	98.00	0.00	0.00	61.90	750.00	88.80	83.00	155.00	1,700.00	0.00	131.00	329.00	4,190.00	3,780
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	549.00	370.00	0.00	0.00	31.70	505.00								6,250.00	6,720
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Lateral 6

Sample description: Lateral 6

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

50 - 11/10/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	7,700.00		12.80
DL	100.00	100.00	100.00		0.01

50 - 06/19/2023: Corn, silage

50 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	600.00	4,800.00		7.90
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	1,000.00	7,500.00		6.80
DL	100.00	100.00	100.00		0.01

70 - 11/07/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage
Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 61.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,500.00	1,000.00	6,200.00		9.16
DL	100.00	100.00	100.00		0.01

70 - 11/07/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,100.00	6,900.00		11.30
DL	100.00	100.00	100.00		0.01

70 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,700.00	500.00	7,000.00		5.30
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 51.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	500.00	5,900.00		5.00
DL	100.00	100.00	100.00		0.01

Corn Shed - 11/08/2022: Oats, silage-soft dough

Corn Shed - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,400.00	1,200.00	8,100.00		12.00
DL	100.00	100.00	100.00		0.01

Corn Shed - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,400.00	10,700.00		8.50
DL	100.00	100.00	100.00		0.01

Hennes - 06/17/2023: Corn, silage

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	7,600.00		10.20
DL	100.00	100.00	100.00		0.01

Hennes - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,000.00	7,500.00		8.60
DL	100.00	100.00	100.00		0.01

Horse Pasture - 11/09/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	800.00	6,100.00		12.30
DL	100.00	100.00	100.00		0.01

Horse Pasture - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	800.00	5,700.00		5.90
DL	100.00	100.00	100.00		1.00

Manuel N/S - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	1,000.00	7,000.00		11.00
DL	100.00	100.00	100.00		0.01

Manuel N/S - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,900.00	600.00	5,200.00		7.50
DL	100.00	100.00	100.00		0.01

Manuel S/S - 11/08/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,300.00	1,200.00	8,200.00		11.00
DL	100.00	100.00	100.00		0.01

Manuel S/S - 06/18/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	900.00	6,200.00		6.30
DL	100.00	100.00	100.00		0.01

Marks Trees - 01/01/2023: Almonds

Almonds

Sample and source description: Almonds

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	65,000.00	11,000.00	70,500.00		0.00
DL	100.00	100.00	100.00		0.01

Morris - 11/09/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	1,000.00	6,000.00		8.29
DL	100.00	100.00	100.00		0.01

Morris - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	800.00	6,200.00		6.70
DL	100.00	100.00	100.00		0.01

Pasture - 11/09/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	900.00	5,700.00		10.20
DL	100.00	100.00	100.00		0.01

Pasture - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	800.00	5,700.00		5.90
DL	100.00	100.00	100.00		0.01

Silveria 1 - 11/09/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,200.00	7,900.00		11.30
DL	100.00	100.00	100.00		0.01

Silveria 1 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	500.00	6,600.00		5.50
DL	100.00	100.00	100.00		0.01

Silveria 2 - 11/09/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,200.00	7,900.00		11.30
DL	100.00	100.00	100.00		0.01

Silveria 2 - 06/20/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	500.00	6,600.00		5.50
DL	100.00	100.00	100.00		0.01

The 60 - 11/07/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,100.00	1,200.00	8,100.00		9.94
DL	100.00	100.00	100.00		0.01

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,200.00	8,500.00		11.00
DL	100.00	100.00	100.00		0.01

The 60 - 06/19/2023: Corn, silage

The 60 - 06/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	800.00	6,100.00		7.20
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,600.00	900.00	6,800.00		7.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

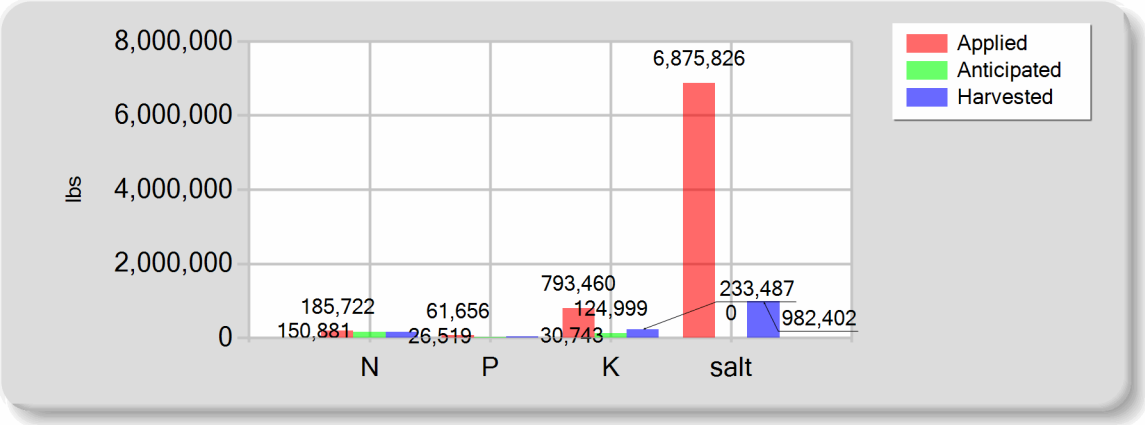
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

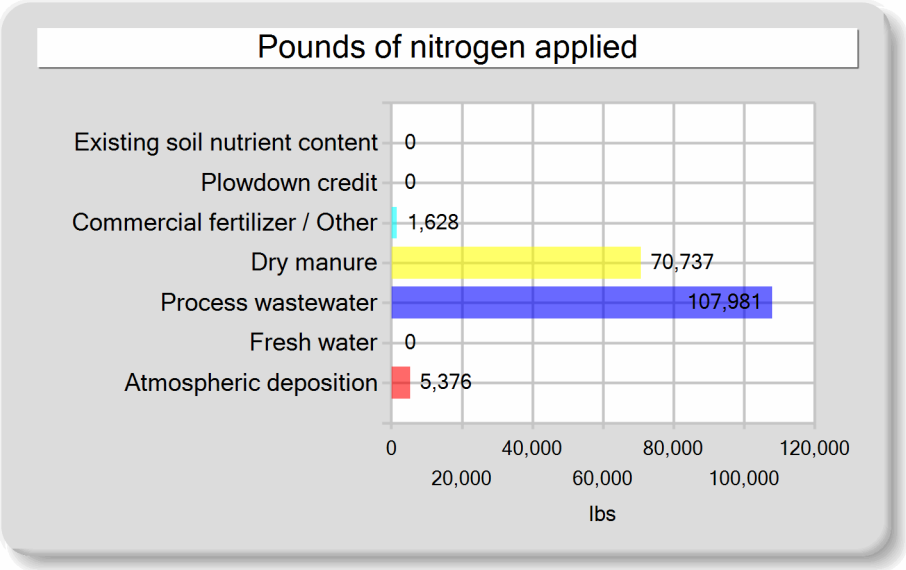
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

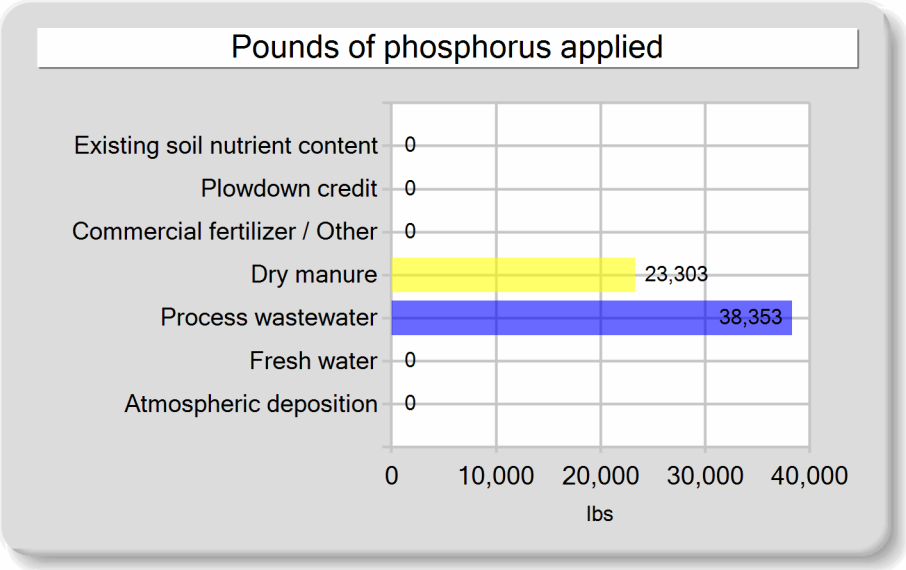
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,628.00	0.00	0.00	0.00
Dry manure	70,737.14	23,303.15	30,025.28	3,219,139.20
Process wastewater	107,980.76	38,352.66	763,435.19	2,689,488.52
Fresh water	0.00	0.00	0.00	967,197.79
Atmospheric deposition	5,376.00	0.00	0.00	0.00
Total nutrients applied	185,721.90	61,655.82	793,460.47	6,875,825.51
Anticipated crop nutrient removal	150,880.80	26,518.60	124,998.80	0.00
Actual crop nutrient removal	159,285.77	30,743.27	233,486.95	982,402.00
Nutrient balance	26,436.13	30,912.54	559,973.52	5,893,423.51
Applied to removed ratio	1.17	2.01	3.40	7.00

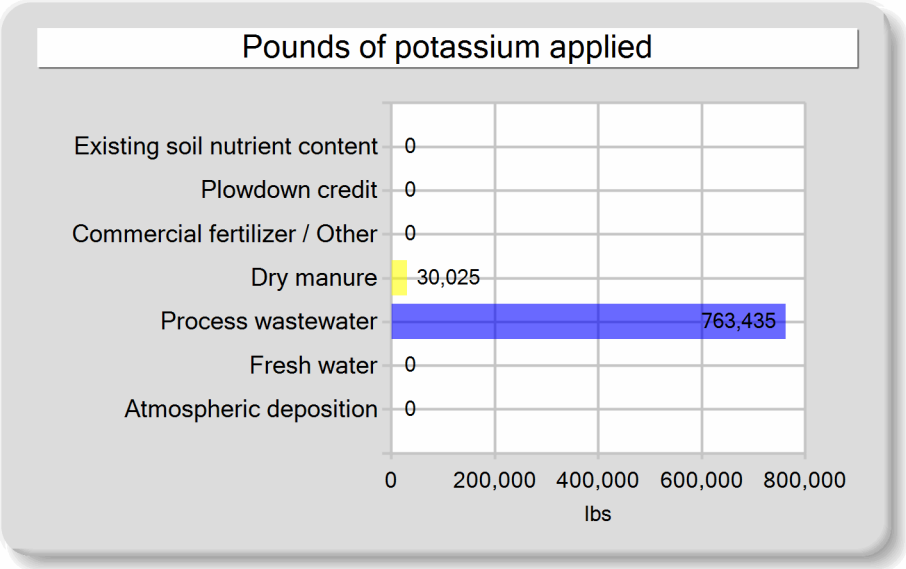
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

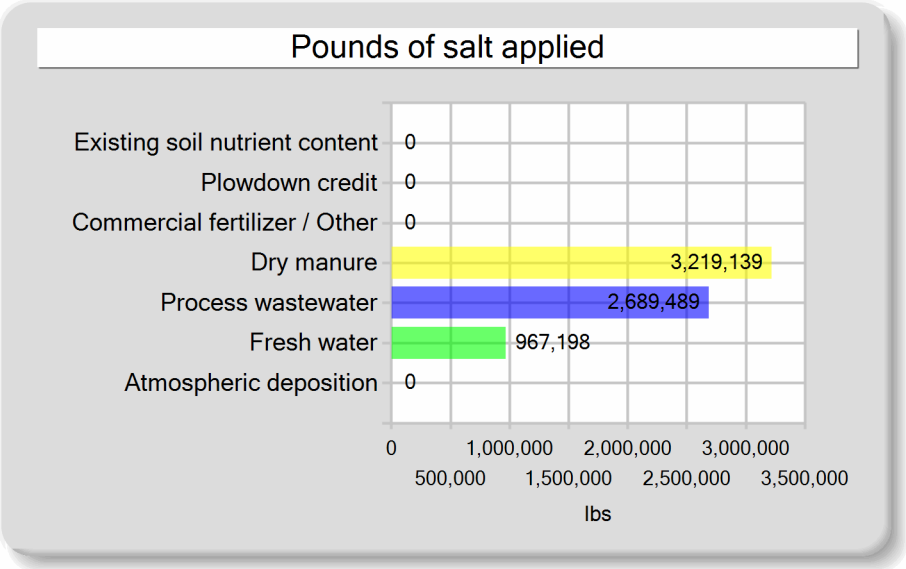


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mark Ahlem Mark Ahlem (Jun 19, 2024 16:12 PDT)	Mark Ahlem Mark Ahlem (Jun 19, 2024 16:12 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Mark Ahlem	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	Jun 19, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Charles Ahlem Ranch*
23546 W. American
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
East Home MB	T3291014A	Electrical Conductivity (EC)	EPA 120.1 and	1.06	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	680	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Charles Ahlem Ranch*
23546 W. American
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W/S Home Barn	T3291014B	Electrical Conductivity (EC)	EPA 120.1 and	1.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.88	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	693	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Charles Ahlem Ranch*
23546 W. American
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Far East Home Barn	T3291014C	Electrical Conductivity (EC)	EPA 120.1 and	1.02	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.549	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	655	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

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Compliance Analysis Report

Customer
Charles Ahlem Ranch*
23546 W. American
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/18/23 9:45 AM
SUBMITTED BY: KJ
ANALYZED DATE: 10/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Hennes	T3291014D	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	686	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	102	101 (T3291008A)	99.4 (T3291008A)	1.20
Electrical Conductivity (EC)	97.6			0.093
Nitrate Nitrogen (NO3-N)	96.9			2.49

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Charles Ahlem
Mailing address: Ranches
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: SB
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID-	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/16	10:45	Domestic	East Home MB			14.8' ✓				
2		11:00	wells	W/S Home Barn			14.0' ✓				
3		11:15		Far East Home Barn			13.6' ✓				
4		11:30		Hennes			14.6' ✓				
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		TRAG				
2nd		DENELE ANALYTICAL, INC.				T3291014

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Mark Ahlem

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:
Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 1/4/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 390,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company/Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 1/30/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,090,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street | City | Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name | Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company/Person:

Number and Street | City | Zip Code

Contact Person:

Name | Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 2/2/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1900,000 Gallons

Method used to determine volume of process wastewater: PD

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

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Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street | City | Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name | Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street | City | Zip Code

Contact Person:

Name | Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 2/15/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 225,000 Gallons

Method used to determine volume of process wastewater: PD

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Mark Ahlem

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 3/10 - 3/12

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 735,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Mark Ahlem

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street | City | Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name | Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street | City | Zip Code

Contact Person:

Name | Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 4/10-4/11

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,290,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company/Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 6/4-6/5

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 100,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 4/10-4/11

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 700,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street | City | Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name | Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street | City | Zip Code

Contact Person:

Name | Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 7/27/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 690,000 Gallons

Method used to determine volume of process wastewater: FO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 9/5-9/6

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,710,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street | City | Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name | Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street | City | Zip Code

Contact Person:

Name | Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 8/15-8/16

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1,900,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Mark Ahlem

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 0/23- 0/24

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 855,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Mark Ahlem

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMA #2 **23970 August Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 9/22 - 9/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 204000 Gallons

Method used to determine volume of process wastewater: RO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMC Jerseys **23204 Bloss Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 8/7/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 405,000 Gallons

Method used to determine volume of process wastewater: PD

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMC Jerseys **23204 Bloss Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 9/7/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 415,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: MARK AHLEM

Name of Dairy Facility: **Charles Ahlem Ranches**

Facility Address: **23546 W. American Ave** **Hilmar, CA 95324**

Number and Street City Zip Code

Contact Person Name and Phone Number: **Mark Ahlem** **(209) 668-0867**

Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person:

Address of Hauling Company /Person:

Number and Street City Zip Code

Contact Person:

Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

CMC Jerseys **23204 Bloss Ave** **Hilmar, CA 95324** **209-668-0867**

Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Same As Above

Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 9/27/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons):

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 304,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____(Operator shall provide initials here to acknowledge this requirement).

Certification:
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Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

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2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lloyd Erlandson (Name) 8658 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-761-3942 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-041 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 4/20/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 300,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lloyd Erlandson (Name) 8658 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-761-3942 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-041 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 4/28/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 720,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company/Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Lloyd Erlandson (Name) 8658 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-761-3942 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-041 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 8/13/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 1900,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-003 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 3/3/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 540,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-180-003 (Assessor's Parcel Number)

Dates Hauled: 4/16/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 300,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-003 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 6/18/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 240,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-003 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 7/10/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 420,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

045-180-003 (Number and Street) _____ (City) _____ (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 9/12/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 300,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: NA Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company/Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact Information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-180-003 (Assessor's Parcel Number)

Dates Hauled: 9/22/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 300,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem
Mark Ahlem (Jun 19, 2024 16:12 PDT) Date: _____

Hauler's Signature: N/A Date: _____

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Mark Ahlem

Name of Dairy Facility: Charles Ahlem Ranches

Facility Address: 23546 W. American Ave (Number and Street) Hilmar (City) 95324 (Zip Code)

Contact Person Name and Phone Number: Mark Ahlem (Name) 209-668-0867 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _____

Address of Hauling Company /Person: _____ (Number and Street) _____ (City) _____ (Zip Code)

Contact Person: _____ (Name) _____ (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Ralph Seward Inc. (Name) 8827 Washington Rd. (Number and Street) Hilmar (City) 95324 (Zip Code) 209-484-8014 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____ (Number and Street) _____ (City) _____ (Zip Code) 045-180-003 (Assessor's Parcel Number)

Dates Hauled: 11/2/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____ Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): _____

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: 400,000 Gallons

Method used to determine volume of process wastewater: PO

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Mark Ahlem (Mark Ahlem (Jun 19, 2024 16:12 PDT)) Date: _____

Hauler's Signature: N/A Date: _____

2023 Annual Report- Charles Ahlem Ranches

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
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"2023 Annual Report- Charles Ahlem Ranches" History

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